

Separate-Absolute Majorant Barrier

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Abstract

The cloud-extracted trace is a cancellation-sensitive residual. We prove that a sectorwise absolute majorant, formed before physical/atomic cancellation, has root rate at least the Hardy-scaled Perron rate $r_H^{-1} > 1$. The obstruction is exact and independent of the cloud size. An audit of 352 archived order cases confirms root rates from 1.2532 to 3.5568 and enormous majorant-to-residual gains. This rules out one cancellation-blind estimate, not signed or grouped quotient loops.

1 The cancellation-blind quantity

Write the scaled trace decomposition at a fixed noise value as

$$\tau_n = t_n - p^n - q^n - \sum_{s \in S} s^n,$$

where $t_n = \text{Tr } A^n$, p is the Perron root, q the parity root, and S the selected cloud. Define the separate-absolute trace majorant

$$U_n^{\text{sep}} = |t_n| + |p|^n + |q|^n + \sum_{s \in S} |s|^n. \quad (1)$$

An absolute periodic-loop integral for the physical sector is no smaller than the first term in (1); hence (1) is the most favorable sectorwise absolute test for the present purpose.

Theorem 1 (Perron lower barrier). *Suppose the scaled Perron root obeys $|p_\sigma| \geq \rho > 1$ on a noise set. Then for every selected cloud and every $n \geq 1$,*

$$U_n^{\text{sep}}(\sigma) \geq \rho^n, \quad \liminf_{n \rightarrow \infty} (U_n^{\text{sep}}(\sigma))^{1/n} \geq \rho.$$

Consequently no estimate of the form $U_n^{\text{sep}}(\sigma) \leq Mq^n$ with $q < 1$ can hold for all sufficiently large n on that set.

Proof. The Perron summand in (1) is nonnegative and equals $|p_\sigma|^n \geq \rho^n$. Taking n th roots and then a liminf proves the claim. A subunit geometric upper bound would force the limsup of the same roots to be at most $q < 1$, a contradiction. $\square$

For the current Hardy scaling $r_H = 0.85$, the archived Perron value is exactly

$$|p_\sigma| = r_H^{-1} = 1.1764705882352942.$$

This is the same normalization that makes the deterministic numerator target in RH-243 nontrivial. The obstruction is therefore structural, not a poor choice of numerical tolerance.

2 Finite audit

Using the RH-222 clouds and RH-236 full/residual trace powers, we evaluated (1) for all 32 endpoints and orders 2–12 [1]. There are 352 cases. The resulting individual root rates $U_n^{\text{sep}})^{1/n}$ lie in

$$1.253165378203787 \leq (U_n^{\text{sep}})^{1/n} \leq 3.5568338003788416,$$

so all 352 exceed one. The cancellation gain $U_n^{\text{sep}}/|\tau_n|$ ranges from 42.68449383956666 to $7.802289221403802 \times 10^{15}$; its maximum is at $(\sigma, \text{side}, n) = (0.005, \text{right}, 10)$.

quantity	value
endpoints/orders	32 / 352
root-rate range	1.253165–3.556834
cases with root rate > 1	352/352
majorant/residual range	42.6845– 7.8023×10^{15}

3 Scope and route consequence

Theorem 1 rules out only estimates that take absolute values of the physical, Perron, parity, and cloud sectors separately. It does not rule out the exact graded cancellation of RH-242, the orthogonal quotient of RH-245, complex kernel cancellations, or a different normalization in a new route. The finite audit uses floating archived roots and orders 2–12; it is not an all-order interval proof.

The viable envelope route must group sectors before majorization. The deterministic numerator anchor remains unresolved. Gate A remains open and Gates B–E are untouched. No Hilbert–Polya operator, zeta-divisor equality, Riemann-zero identification, or RH implication is claimed.

References

- [1] Bin Wang. Cloud-extracted trace moment atlas. 2026. RH-236 preprint.